

PCA in face recognition systems



PCA to recognize the image.

I. Training Phase

Suppose we have four 2×2 images. These are training face images

$$I_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$I_4 = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

Step 1: Each image is converted into a column vector.
that is,

- flatten row wise

$$I_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

$$I_4 = \begin{bmatrix} 4 \\ 5 \\ 6 \\ 7 \end{bmatrix}$$

Note that every face is now represented as 4×1 column vector.

Step 2: compute the mean face

$$\text{mean}(I_1) = \frac{1+2+3+4}{4} = 2.5$$

$$\text{mean}(I_2) = \frac{2+3+4+5}{4} = 3.5$$

$$\text{mean}(I_3) = \frac{3+4+5+6}{4} = 4.5$$

$$\text{mean}(I_4) = \frac{4+5+6+7}{4} = 5.5$$

Therefore $\mu = \begin{bmatrix} 2.5 \\ 3.5 \\ 4.5 \\ 5.5 \end{bmatrix}$

Step 3: Normalize (subtract the mean)

$\phi_1 = I_1 - \mu$ and so on.

$$\phi_1 = \begin{bmatrix} -1.5 \\ -1.5 \\ -1.5 \\ -1.5 \end{bmatrix} \quad \phi_2 = \begin{bmatrix} -0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{bmatrix} \quad \phi_3 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} \quad \phi_4 = \begin{bmatrix} 1.5 \\ 1.5 \\ 1.5 \\ 1.5 \end{bmatrix}$$

Arrange normalized vectors as columns.

$$X_c = \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix}$$

Step 4: Covariance matrix (C) (Refer notes)

PCA computes $C = \frac{1}{n-1} X_c^T \cdot X_c$

This 'C' is used for computing eigen values and Eigen vectors. using $\det(C - \lambda I) = 0$
 λ is Eigen values.

Step 5: Suppose the Covariance matrix (C) gives the following Eigen values: $[18, 2, 0.4, 0.1]$

Choose The largest Eigen values: 18, & 2 (and 0.4, and 0.1 both are less than 1)

Since the first two eigen values explain almost all the variance, we ~~keep~~ only eigen vectors of these two.

Let's assume,

Eigen vector of eigen value '1' is = $\begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$
(v_1)

Eigen vector (v_2) of Eigenvalue '2' is = $\begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{bmatrix}$

Step 6: Compute the weights of every training face image.

formula $w = v_i \cdot \phi_i$ [Each face image gets one weight for each eigen vector]

Face image 1 (I_1):

$$\text{weight on eigen value '1'} \Rightarrow w_1 = 0.5 \times (-1.5) + 0.5 \times (-1.5) + 0.5 \times (-1.5) + 0.5 \times (-1.5)$$

$$\therefore w_1 = -3$$

$$\text{Weight on eigen value '2'} \Rightarrow 0.5 \times (-1.5) + (-0.5) \times (-1.5) + 0.5 \times (-1.5) + (-0.5) \times (-1.5)$$

$$= -0.75 + 0.75 - 0.75 + 0.75 = 0$$

face 1 weights are = $(-3, 0)$

Similarly, find weights of face image 2, 3 & 4.

for face image 2 (I_2): weight is $(-1, 0)$

for face image 3 (I_3), the weight is $(1, 0)$

for face image 4 (I_4), the weight is $(3, 0)$

This is the face database.

I. Recognition Phase:

$$\text{Test face image} = \begin{bmatrix} 2.2 & 3.2 \\ 4.2 & 5.2 \end{bmatrix}$$

$$\text{convert it into vector} = \begin{bmatrix} 2.2 \\ 3.2 \\ 4.2 \\ 5.2 \end{bmatrix}$$

Normalize (subtract mean)

$$\text{we know that the mean } \mu \text{ is } \begin{bmatrix} 2.5 \\ 3.5 \\ 4.5 \\ 5.5 \end{bmatrix}$$

$$\text{subtract the mean, } \phi = \begin{bmatrix} -0.3 \\ -0.3 \\ -0.3 \\ -0.3 \end{bmatrix}$$

Compute weights:

for eigen value 18,

$$\begin{aligned} w_1 &= 0.5 \times (-0.3) + 0.5 \times (-0.3) + \\ &\quad 0.5 \times (-0.3) + 0.5 \times (-0.3) \\ &= 0.6 \end{aligned}$$

for eigen value 2,

weight is 0.

weight vector of testing face image is
 $(-0.6, 0)$

$\Rightarrow$ compare it with training face database,
find Euclidean distance between these two.

We have,

face database (weights)

	weights
I_1	$(-3, 0) (x_1, y_1)$
I_2	$(-1, 0)$
I_3	$(1, 0)$
I_4	$(3, 0)$

testing face image
weight x_2, y_2
 $(-0.6, 0)$

Euc. distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Distance to $I_1 = \sqrt{(-0.6 + 3)^2} = 2.4$

Distance to $I_2 = \sqrt{(-0.6 + 1)^2} = \textcircled{0.4}$ smallest distance

Distance to $I_3 = \sqrt{(-0.6 - 1)^2} = 1.6$

Distance to $I_4 = \sqrt{(-0.6 - 3)^2} = 3.6$

Decision: The smallest distance is 0.4.

Therefore, New face (Testing face) is very close to training face 2.